CMFNet: Cross Mamba Fusion Network for Hyperspectral and LiDAR Data Classification

Ziqi Li, Jiang Wu, Yonghong Zhang, and Yu Yan

***Abstract*—The joint classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data, which leverages their complementary information, has emerged as a crucial research direction in remote sensing. However, the significant differences in imaging mechanisms result in high heterogeneity between HSI and LiDAR data in terms of dimensionality and feature distributions, posing substantial challenges for semantic representation and correlation of multimodal data. To address this challenge, this article proposes a cross Mamba fusion network (CMFNet) for hyperspectral and LiDAR data classification. The proposed framework adopts a dual-branch architecture to facilitate multimodal feature extraction and interaction, initially performing multiscale convolutional feature embedding before adaptively fusing global and local features through a dedicated global–local feature extraction (GLFE) module. To further exploit the complementary advantages of multisource data, we design a cross Mamba fusion (CMF) module that extends the Mamba architecture to enable efficient dual-modal interaction. This module enhances cross-modal feature fusion while reducing computational redundancy and achieves hierarchical feature extraction through a multilayer structure. Comprehensive experiments on three benchmark remote sensing datasets demonstrate that CMFNet significantly outperforms state-of-the-art methods, with ablation studies thoroughly validating the effectiveness of each component. Habitually, the CMFNet's source code can be made publicly accessible on my profile page at https://github.com/li-zi-qi/CMFNet**

***Index Terms*—Classification, hyperspectral image (HSI), light detection and ranging (LiDAR), Mamba, multimodal.**

I. Introduction

With the rapid advancement of remote sensing technology, hyperspectral image (HSI) and light detection and ranging (LiDAR) have emerged as two cornerstone technologies in Earth observation [1], [2], [3], [4]. These complementary techniques offer distinct advantages in spatial resolution, spectral information acquisition, and 3-D structural characterization. Their applications span a wide range of fields, including land cover classification [5], target detection [6], environmental monitoring [7], urban planning [8], [9], precision agriculture [10], [11], and food security [12].

HSI acquired through continuous narrowband imaging provides rich spectral information capable of precisely characterizing the chemical composition of surface materials [13], [14]. In contrast, LiDAR uses active laser pulses to capture 3-D structural features of targets, such as eleva tion and surface texture [15]. However, single-sensor data exhibit inherent limitations in complex scene classification: hyperspectral imagery is susceptible to illumination variations, shadows, and the "same object with different spectra" phenomenon, while LiDAR lacks spectral characterization capabilities, making it difficult to distinguish between spectrally similar but materially distinct features. Consequently, the synergistic exploitation of multisource remote sensing data to extract integrated spectral–spatial features has become a pivotal challenge in improving land cover classification accuracy [16], [17], [18].

With the rise of deep learning methods, HSI and LiDAR classification algorithms based on convolutional neural networks (CNNs) have emerged [19]. Unlike traditional machine learning approaches that rely on manually designed features, deep learning models can automatically extract multilevel abstract features from data, thereby more effectively capturing the complex nonlinear relationships between HSI and LiDAR data [20], [21], [22], [23]. Xu et al. [24] proposed a dual-branch CNN framework, demonstrating that separately extracting HSI spectral features and LiDAR 3-D structural features through heterogeneous networks can significantly improve classification accuracy in complex scenes. Hang et al. [25] introduced an efficient CNN framework using two coupled CNNs to fuse HSI and LiDAR data—one CNN learns spectral–spatial features from HSI data, while the other captures elevation information from LiDAR data, using both feature-level and decision-level fusion methods to fully integrate these heterogeneous features. Due to the limitations of 2-D convolution in feature extraction, He et al. [26] proposed a multiscale 3-D CNN (M3D-CNN), which jointly learns 2-D multiscale spatial features and 1-D spectral features from HSI data in an end-to-end manner. Ben Hamida et al. [27] proposed a novel 3-D-CNN architecture that leverages 3-D convolution

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to process both spatial and spectral information, delivering better classification results with fewer trainable parameters while maximizing the information provided by the data and fundamentally reducing computational costs. Zhao et al. [28] introduced a hierarchical random walk network (HRWN), designing parallel convolutional networks to separately extract spectral and spatial features, where the prediction distribution of a dual-tunnel CNN serves as a global prior while pixel affinity reflects local similarity, thereby enhancing the spatial consistency of deep networks for joint classification of HSI and LiDAR data.

While CNNs can extract contextual information in the spatial domain, they struggle to capture spatial–spectral interactive information, particularly when processing large-scale, multiscale data and capturing long-range dependencies [29], [30]. As an effective backbone network, the transformer [31] leverages self-attention mechanisms to process and analyze sequential data more efficiently. Whether for unimodal feature extraction or multimodal heterogeneous feature fusion, transformers exhibit strong global modeling capabilities in remote sensing image classification. Hong et al. [32] proposed a spectral transformer (ST) model that learns spectral representations from grouped neighboring bands and incorporates a cross-layer transformer encoder (TE) module, though these methods only use transformers for spectral information processing without using spatial information. To address this, Sun et al. [33] introduced a spectral–spatial feature tokenization transformer (SSFTT) method to extract spectral–spatial features and high-level semantic features, while Roy et al. [34] developed a mathematical morphology transformer (MMT) implementing learnable spectral–spatial morphological networks that combine mathematical morphological convolution with attention mechanisms to enhance structural and shape information interactions. Yang et al. [35] proposed a novel transformer-based cross-attention framework that leverages LiDAR data to intelligently select optimal HSI bands, effectively addressing spectral redundancy while significantly boosting multisource classification performance. Song et al. [36] proposed a novel height information-guided hierarchical fusion-separation network (HFSNet) for joint classification of HSI and LiDAR data. Chang et al. [37] proposed a novel dual-branch transformer-based framework (DSymFuser), which incorporates a symmetric fusion transformer (SFT) to enhance cross-modal correlation mining and fully exploit complementary cues from heterogeneous modalities, thereby improving classification performance. Li et al. [38] proposed a feature reconstruction-guided fusion network (FRGFNet), which uses feature decomposition to reconstruct HSI features and introduces a cross-modal attention mechanism to effectively fuse HSI and LiDAR data, significantly improving semantic alignment and classification performance. Gao et al. [39] proposed a prototype-based information compensation network (PICNet), which achieves effective complementary fusion and land cover classification of hyperspectral and SAR/LiDAR data through frequency interaction and cross-modal prototype guidance mechanisms.

Despite significant progress in HSI and LiDAR data classification using both the CNN-based and transformer-based approaches [40], [41], [42], [43], the inherent quadratic computational complexity of transformer models significantly increases computational costs. Therefore, an enhanced state-space model (SSM) called Mamba network [44], incorporating a selective scanning mechanism, has emerged as a highly promising solution for computer vision and remote sensing image analysis [45], [46], [47], [48], [49]. Unlike transformer architectures, the Mamba network incorporates a selective state-space mechanism that achieves linear computational complexity with sequence length, offering dramatic efficiency improvements for processing long sequences. Liao et al. [50] proposed a Mamba-based joint classification network for HSI and LiDAR data (HLMamba), constructing a multimodal Mamba fusion module (MMFM) that effectively models long-range dependencies in multimodal sequences to enhance fusion performance. Gao et al. [51] developed a multiscale feature fusion Mamba (MSFMamba) network, which effectively extracts feature information from multimodal data by constructing spectral, spatial, and fusion Mamba modules, thereby improving network performance. Cao et al. [52] proposed a CLIP-driven multimodal fusion model (M^3amba), which incorporated modality-specific adapters and a linear-complexity Mamba architecture to enhance semantic understanding and improve the efficiency and consistency of cross-modal feature fusion. While Mamba models have shown promising performance potential in multisource data fusion, several critical challenges remain to be addressed.

1) *Complexity of Multisource Heterogeneous Data Fusion:* Requires addressing the heterogeneity in spatial resolution and imaging mechanisms between different sensors to achieve effective cross-scale fusion of local features and global contextual information.
2) *Tradeoff Between Model Efficiency and Performance:* There is an urgent need to design computationally efficient novel network architectures that optimize computational resource consumption while maintaining feature representation capability for multimodal remote sensing data.
3) *Optimization of Cross-Modal Feature Interaction:* It is necessary to establish an effective cross-modal feature interaction mechanism to fully exploit the complementary information between HSI and LiDAR data.

To overcome these limitations, this article proposes a cross Mamba fusion network (CMFNet) for hyperspectral and LiDAR data classification. The framework uses a dual-branch structure for feature extraction and interaction between the two modalities, where each modal branch contains a series of global–local feature extraction (GLFE) modules designed to extract hierarchical and discriminative unimodal features. Within each GLFE module, we first use CNN and Mamba networks to extract global and local features, and then use two depthwise separable convolutions to adaptively aggregate global and local information. Furthermore, to fully explore interactions between multimodal features while minimizing computational redundancy, we design a cross Mamba fusion (CMF) module for multisource feature fusion. This extends

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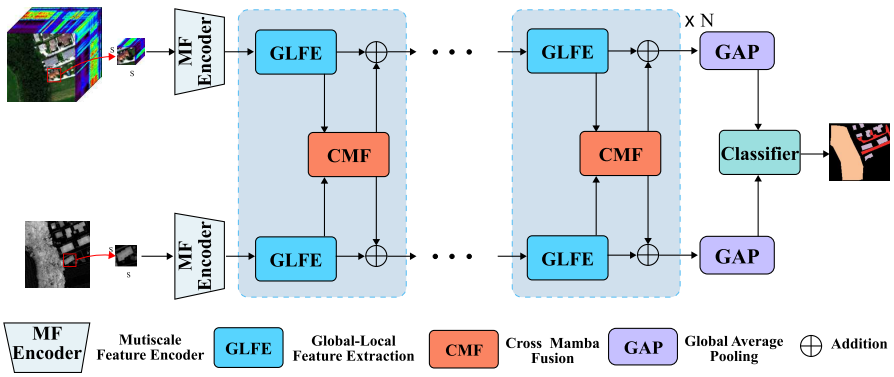


Fig. 1. Overall framework of the proposed CMFNet for HSI and LiDAR classification. First, the MFEncoder extracts preliminary features from the input HSI and LiDAR patches. Then, the GLFE module is used to simultaneously capture both global and local information. Next, the proposed CMF modules perform deep-level bidirectional feature interactions between modalities. These fusion modules are stacked in a cascaded manner (×N) to progressively refine cross-modal representations. Finally, the features are aggregated through GAP and fed into a classifier for pixelwise prediction.

the original Mamba to accommodate dual inputs and enhances cross-modal feature interactions, executed after each GLFE module to mine cross-modal relationships at different hierarchical levels. Our main contributions can be summarized as follows.

1) We propose CMFNet, a novel Mamba-based classification framework that fully leverages the capabilities of Mamba networks to achieve GLFE and cross-modal feature interaction through a multilevel architecture.
2) We develop a GLFE module that innovatively combines Mamba networks with lightweight CNNs to adaptively integrate intramodal global and local features. The multilayer design further enables hierarchical multimodal global–local feature fusion.
3) We creatively design a CMF module for cross-modal interaction mining, which significantly enhances feature interaction between multisource data through cross Mamba operations, maximizing the complementary advantages of both the modalities.
4) Extensive experiments conducted on three benchmark multimodal remote sensing classification datasets demonstrate that our approach outperforms existing state-of-the-art methods. Ablation studies further validate the effectiveness of each key component in our proposed framework.

II. Methodology

This section presents in detail our CMFNet for hyperspectral and LiDAR data classification. As illustrated in Fig. 1, the proposed framework adopts a dual-branch architecture for multimodal feature extraction and interaction, consisting of three key components: multiscale feature encoder (MFEncoder), GLFE, and CMF. The network first extracts multiscale shallow features from both the modalities through the encoder. These features are then progressively processed through a series of GLFE modules, each followed by a CMF module that effectively captures cross-modal interactions. The fused features subsequently serve as input to the next-level GLFE module, forming a hierarchical feature learning framework that gradually transforms low-level visual features into high-level semantic representations. This multistage architecture enables progressive fusion of multimodal information from elementary to advanced levels, thoroughly exploiting complementary information across modalities. Finally, the features extracted through multistage feature learning and cross-modal interaction undergo additive fusion before being fed into a fully connected layer for the classification task. The hierarchical design ensures comprehensive mining of cross-modal correlations while maintaining the distinct characteristics of each data modality throughout the fusion process.

A. Preliminaries

1) State-Space Model: SSM represents a continuous-time system model whose core mechanism achieves nonlinear mapping from a 1-D input signal $x(t) \in \mathbb{R}^N$, to a 1-D output signal $y(t) \in \mathbb{R}^N$, through a hidden state vector $h(t) \in \mathbb{R}^N$. The dynamic characteristics of this system are typically mathematically described using ordinary differential equation (ODE), with the fundamental formulation expressed as

$$\begin{cases} h'(t) = Ah(t) + Bx(t) \\ y(t) = Ch(t) \end{cases} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ is the state transition matrix, and $B \in \mathbb{R}^{N\times 1}$ and $C \in \mathbb{R}^{N\times 1}$ are the mapping matrices. Equation (1) reveals the global receptive field property of SSM, where the current output $y(t)$ integrates the cumulative effects of all historical inputs through continuous-time system dynamics. Under the condition of constant system parameters A, B, and C, the equation reduces to a linear time-invariant (LTI) system. However, when these system parameters vary dynamically over time, the equation describes a more expressive linear time-varying (LTV) system, which constitutes the fundamental modeling approach adopted by Mamba. Essentially, while LTI systems maintain fixed parameter structures incapable of adaptively adjusting their response characteristics based on input content, LTV systems introduce a time-varying parameter mechanism that enables dynamic perception and selective response to input stimuli. This crucial distinction endows LTV systems with superior performance in sequence modeling tasks, as they can effectively capture and adapt to the evolving patterns in sequential data.

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2) Discretization: For effective implementation of SSM in deep learning applications, the continuous-time system must undergo discretization. We introduce a crucial timescale parameter Δ to transform the continuous parameters A and B into their discrete counterparts $\bar{A}$ and $\bar{B}$. This conversion is achieved through the zero-order hold (ZOH) method, with the discretization formulas given by

$$\begin{cases} \bar{A} = \exp(\Delta A) \\ \bar{B} = (\Delta A)^{-1}\left(\exp(\Delta A) - I\right) \cdot \Delta B \\ \quad \approx (\Delta A)^{-1}(\Delta A)\,\Delta B \\ \quad = \Delta B. \end{cases} \tag{2}$$

In the above equation, Δ is the time scale parameter. After discretization, the SSM discrete system can be represented as

$$\begin{cases} h_t = \bar{A}h_{t-1} + \bar{B}x_t \\ y_t = Ch_t. \end{cases} \tag{3}$$

To enhance computational efficiency, convolution is typically used to accelerate the linear recursive process described above. Ultimately, this process can be represented as follows:

$$\begin{cases} \bar{K} = \left(C\bar{B}, C\overline{AB}, \ldots, C\bar{A}^{L-1}\bar{B}\right) \\ y = x * \bar{K} \end{cases} \tag{4}$$

where L is the length of the input sequence and $\bar{K} \in \mathbb{R}^L$ is the structured convolution kernel.

3) Selective Scanning Mechanism: SSM relies heavily on the simplified LTI assumption, which offers the advantage of linear time complexity but fails to effectively capture contextual information within input sequences. To address this limitation, Mamba introduces a selection mechanism and proposes a novel Selective SSM (S6) that enables dynamic interactions between sequence states. The S6 model incorporates three key hardware-aware optimizations: 1) kernel fusion; 2) parallel scanning; and 3) recomputation techniques. This selective scanning architecture allows Mamba to achieve remarkable computational speed while maintaining relatively low memory requirements. The input–output relationship between x and y can be formally expressed as

$$y = \text{Mamba}(x) \tag{5}$$

where Mamba$(\cdot)$ is the Mamba function.

B. Multiscale Feature Encoder

With its hierarchical feature extraction mechanism, the CNN progressively compresses the spatial dimensions of input data through stacked multilayer convolution and pooling operations, enabling multiscale feature representation of images. This unique architectural advantage makes CNN particularly suitable for HSI and LiDAR classification tasks, as they can effectively integrate spectral characteristics with spatial contextual information. Accordingly, we use a combination of multiscale 3-D and 2-D convolutions to perform shallow feature extraction from both HSI and LiDAR data, serving as feature embeddings for subsequent processing modules.

We denote the HSI as $X_{\text{HSI}} \in \mathbb{R}^{H\times W\times B}$ and the LiDAR image covering the same area on the Earth's surface as $X_{\text{LiDAR}} \in \mathbb{R}^{H\times W}$, where H and W are the spatial sizes and B is the number of spectral bands. With each pixel point (i,j) of the two modal data, $i \in \{1,2,\ldots,W\}$, $j \in \{1,2,\ldots,H\}$ as the center, patches $X_H^p \in \mathbb{R}^{s\times s\times B}$ and $X_L^p \in \mathbb{R}^{s\times s}$ of the same size are extracted from X_{HSI} and X_{LiDAR}, respectively, $s \times s$ denoting the patch size, where each pixel point represents a category. The label $y_i \in \{1,2,\ldots,K\}$, where K denotes the number of label categories.

The MFEncoder uses a hybrid architecture combining 3-D and 2-D convolutional operations to extract joint spatial–spectral features. The processing pipeline begins by feeding HSI patches from the training set into a 3-D convolutional layer with a kernel size of $8@9 \times 3 \times 3$, where symmetric padding of size $(0 \times 1 \times 1)$ is applied to maintain spatial dimensions. The multiscale 2-D convolution module consists of three parallel 2-D convolution layers after the 3-D convolution layer. Two 2-D convolution layers are used for grouped convolution, and the other is used for point-by-point convolution. The module uses three kernels of different sizes to extract information at multiple scales. The features obtained from these three convolutions are finally summed in an elementwise manner $(\oplus)$ to obtain the HSI features F_H

$$\hat{X}_H^p = \text{Reshape}\left(\text{Conv3D}\left(X_H^p\right)\right) \tag{6}$$

$$\begin{aligned} F_H = {} & \text{Conv2D}\left(\hat{X}_H^p, k_1, g_1, p_1\right) \\ & \oplus\ \text{Conv2D}\left(\hat{X}_H^p, k_2, g_2, p_2\right) \\ & \oplus\ \text{Conv2D}\left(\hat{X}_H^p, k_3, g_3, p_3\right) \end{aligned} \tag{7}$$

where $k_1 = 1$, $g_1 = 1$, $p_1 = 0$, $k_2 = 3$, $g_2 = 4$, $p_2 = 1$, $k_3 = 5$, $g_3 = 6$, and $p_3 = 2$.

Through the aforementioned convolutional operations, we obtain multiscale feature representations of the HSI data.

Similarly, for LiDAR data processing, we use a convolutional kernel of size $64@5 \times 3$ with (2×1) padding in the multiscale 2-D convolution module. This branch adopts a network architecture analogous to the HSI pathway to enhance feature extraction. The output feature map F_L from the second pathway can be formally expressed as

$$\hat{X}_L^p = \text{Reshape}\left(\text{Conv2D}\left(X_L^p\right)\right) \tag{8}$$

$$\begin{aligned} F_L = {} & \text{Conv2D}\left(\hat{X}_L^p, k_1, g_1, p_1\right) \\ & \oplus \text{Conv2D}\left(\hat{X}_L^p, k_2, g_2, p_2\right) \\ & \oplus \text{Conv2D}\left(\hat{X}_L^p, k_3, g_3, p_3\right) \end{aligned} \tag{9}$$

where $k_1 = 1$, $g_1 = 1$, $p_1 = 0$, $k_2 = 3$, $g_2 = 4$, $p_2 = 1$, $k_3 = 5$, $g_3 = 6$, and $p_3 = 2$. Furthermore, batch normalization (BN) and rectified linear unit (ReLU) activation layers are incorporated after each convolutional layer to accelerate model training convergence and enhance feature discriminability.

C. GLFE Module

The MFEncoder demonstrates effective feature extraction for both the modalities but fails to capture long-range dependencies. Given that CNNs excel at extracting local patterns while Mamba networks show remarkable capability in modeling global contextual relationships, we propose a novel architecture as shown in Fig. 2 that strategically combines

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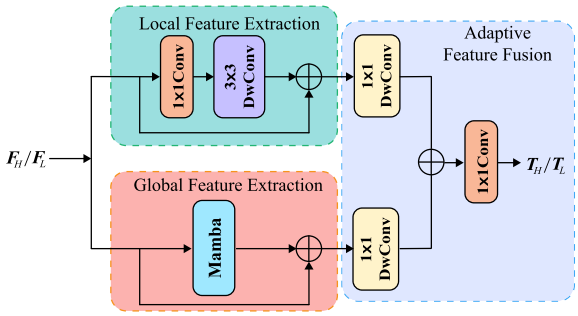


Fig. 2. Overall framework of the proposed GLFE.

CNN and Mamba networks. This design comprehensively captures both global and local feature interactions and uses 1-D convolutions to dynamically fuse these complementary information streams. The feature extraction capability can be quantified as follows. Let C_{in} and C_{out} represent the input and output feature channels, respectively, with a convolution kernel size of $D_k \times D_k$. The computational cost of standard convolution is $D_k^2 \times C_{\text{in}} \times C_{\text{out}}$, while separable convolution reduces this to $D_k^2 \times C_{\text{in}} + C_{\text{in}} \times C_{\text{out}}$. This formulation demonstrates that separable convolution significantly decreases computational load when either the kernel size D_k or input channels C_{in} becomes large. By adopting this separable convolution architecture, we substantially improve the network's parameter efficiency. Specifically, for hyperspectral and LiDAR features F_H and F_L, we apply separable convolution with residual connections to obtain local features F^H_{Local} and F^L_{Local}

$$F^H_{\text{Local}} = \text{DwConv}_{3\times3}\left(\text{Conv}_{1\times1}\left(F_H\right)\right) + F_H \tag{10}$$

$$F^L_{\text{Local}} = \text{DwConv}_{3\times3}\left(\text{Conv}_{1\times1}\left(F_L\right)\right) + F_L \tag{11}$$

where $\text{Conv}_{1\times1}$ denotes pointwise convolution, and $\text{DwConv}_{3\times3}$ represents depthwise convolution with a 3×3 kernel size.

In the global feature extraction stage, we use Mamba to model long-range dependencies for the input features F_H and F_L. The global feature representation can be formulated as

$$F^H_{\text{Global}} = \text{Mamba}\left(F_H\right) + F_H \tag{12}$$

$$F^L_{\text{Global}} = \text{Mamba}\left(F_L\right) + F_L \tag{13}$$

where Mamba$(\cdot)$ denotes the Mamba SSM operation.

The adaptive feature fusion module is designed to dynamically integrate global and local features in an optimized manner. Departing from conventional approaches that use direct summation of global and local features, our proposed fusion mechanism introduces learnable weighting parameters to achieve context-aware feature integration. The mathematical formulation of this adaptive fusion process is expressed as

$$T_H = \text{Conv}_{1\times1}\left(\text{DwConv}_{1\times1}\left(F^H_{\text{Global}}\right) + \text{DwConv}_{1\times1}\left(F^H_{\text{Local}}\right)\right) \tag{14}$$

$$T_L = \text{Conv}_{1\times1}\left(\text{DwConv}_{1\times1}\left(F^L_{\text{Global}}\right) + \text{DwConv}_{1\times1}\left(F^L_{\text{Local}}\right)\right) \tag{15}$$

where $\text{DwConv}_{1\times1}$ denotes a depthwise separable convolution with 1×1 kernels. Specifically, each convolutional layer is sequentially followed by a BN layer and a Gaussian error linear unit (GeLU) activation function. This architecture uses a learnable weighted summation mechanism to achieve channelwise adaptive feature fusion between the two branch outputs, thereby generating dynamically aggregated features with enhanced representational capacity.

From a theoretical perspective, the GLFE module offers clear advantages over existing fusion mechanisms such as self-attention and cross-attention. Unlike conventional attention-based models that entangle local and global representations within a unified attention map, GLFE explicitly separates the modeling of local spatial structures and global contextual dependencies through parallel branches. This decoupled design not only improves learning efficiency but also enhances feature specialization. In addition, by replacing the quadratic-complexity attention mechanism with the linear-complexity Mamba block for global modeling, GLFE significantly reduces computational cost while preserving long-range information.

D. CMF Module

Cross-modal feature interaction plays a vital role in multimodal remote sensing scene classification, where the effective integration of complementary information across modalities can significantly enhance classification performance. However, conventional SSMs are typically designed for unimodal data and often fall short in capturing the complex relationships required for cross-modal fusion. To address this limitation, we propose a CMF module specifically designed to facilitate effective feature integration and interaction between modalities.

As illustrated in Fig. 3, the CMF module first normalizes the input dual-modal features and applies convolutional structures to extract fine-grained feature representations, denoted as T_{Hi} and T_{Li}. To enhance spatial and semantic modeling capabilities, a selective scanning mechanism based on state-space modeling is introduced. This mechanism performs bidirectional scanning along both the spatial and channel dimensions, and the resulting outputs are concatenated to strengthen the contextual representation across regions and channels. In the feature projection phase, a cross-modal driven strategy is adopted: the feature representation T_{Hi} guides the state modeling of T_{Li}, and vice versa. This bidirectional guidance facilitates adaptive interaction between modalities. The fused features are then output as enhanced, modality-aware representations, denoted as T_{Ho} and T_{Lo}.

The detailed implementation is provided in Algorithm 1. In this formulation, the state transition matrix A and skip connection D are initialized as fixed parameters, while the input projection B, output projection C, and timescale parameter $\Delta \in \mathbb{R}$ are dynamically generated through linear transformations of features from the complementary modality. Unlike conventional Mamba architectures that generate parameters solely based on the current sequence, the CMF module introduces a cross-modal parameter transfer mechanism. This design enables the computation of parameters in one modality to be entirely dependent on the projected representations from the other modality, thereby enhancing the coordination and mutual understanding between modalities.

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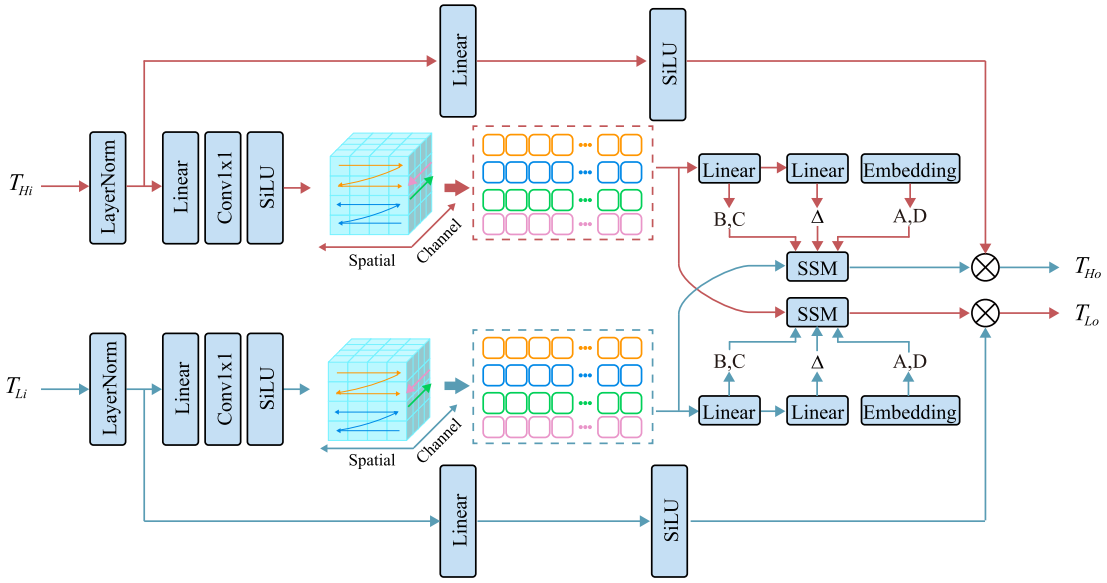


Fig. 3. Overall framework of the proposed CMF. The CMF module extracts fine-grained features from dual-modal inputs and enhances their spatial and channelwise dependencies via an SSM. Bidirectional scanning is performed on both the spatial and channel dimensions, and adaptive cross-modal interactions guide the final fusion process, producing modality-aware representations T_{Ho} and T_{Lo}.

Algorithm 1 Computation of $\mathbf{T}_{Ho}$ in Fig. 3

Require: $\mathbf{T}_{Hi}$, $\mathbf{T}_{Li}$: (L, C), (L, C)
Ensure: $\mathbf{T}_{Ho}$: (L, C)
$\mathbf{A} : (C, N) \leftarrow$ Parameter
$\mathbf{D} : (C) \leftarrow$ Parameter
/* A and D are randomly initialized */
$\mathbf{B} : (L, N) \leftarrow \mathrm{Linear}(\mathbf{T}_{Li})$
$\mathbf{C} : (L, N) \leftarrow \mathrm{Linear}(\mathbf{T}_{Li})$
$\Delta : (L, C) \leftarrow \log(1 + \exp(\mathrm{Linear}(\mathbf{T}_{Li}) + \mathrm{Parameter})$
/* The selection mechanism */
$\overline{\mathbf{A}} : (L, C, N) \leftarrow \exp(\Delta \otimes \mathbf{A})$
$\overline{\mathbf{B}} : (L, C, N) \leftarrow \Delta \otimes \mathbf{B}$
$\mathbf{T}_{Ho} \leftarrow \mathrm{SSM}(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C}, \mathbf{D})(\mathbf{T}_{Hi})$
/* SSM represents 3 implemented using selective scan */
return $\mathbf{T}_{Ho}$

Comparison to MSFMamba: Although MSFMamba also introduces an Fus-Mamba Block based on the SSM to capture intermodal dependencies, its design restricts scanning operations to the spatial dimension only, focusing primarily on modeling spatial correlations in the input sequences. In contrast, the proposed CMF module advances this concept by incorporating a bidirectional selective scanning mechanism along both the spatial and channel dimensions. This dual-dimensional scanning strategy significantly enhances the model's ability to capture complementary information across modalities, enabling it to model both the spatial structures and semantic feature dependencies more comprehensively.

E. Classification Head

After processing through N iterations of GLFE and CMF modules, we obtain enhanced feature representations T_H^N and T_L^N. These features undergo global average pooling (GAP) to produce the final discriminative feature vector T

$$T = \mathrm{GAP}\left(T_H^N\right) \oplus \mathrm{GAP}\left(T_L^N\right) \tag{16}$$

where GAP denotes global average pooling. Subsequently, the fused features are transformed into class probabilities y through a fully connected layer and Softmax function

$$y = \mathrm{Soft\,max}\left(\mathrm{Linear}\left(T\right)\right) \tag{17}$$

where $\mathrm{Linear}(\cdot)$ denotes the weight matrix of the fully connected layer. The entire network is trained using the cross-entropy loss function

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{M}\sum_{i=1}^{M} y_i^T \log\left(\hat{y}_i\right) \tag{18}$$

where M denotes the number of training samples, $y_i \in \mathbb{R}^K$ represents the ground-truth label vector for the ith sample, and $\hat{y}_i \in \mathbb{R}^K$ indicates the predicted label distribution for the ith sample.

III. Experiments and Analysis

To comprehensively validate the effectiveness of CMFNet, we conducted extensive experiments on multiple HSI-LiDAR benchmark datasets. Section III-A details the experimental datasets. Section III-B elaborates on the implementation configurations and evaluation metrics. Section III-C presents thorough parameter analyses. Section III-D demonstrates systematic comparisons against state-of-the-art methods. Section III-E conducts the ablation study to verify each module's contribution.

A. Data Descriptions

1) Trento: The dataset was collected in the rural area of Trento, southern Italy, and includes both HSI and LiDAR

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS FOR THE TRENTO, HOUSTON2013, AND MUUFL DATASETS

NO.	Trento dataset			Houston2013 dataset			MUUFL dataset		
	Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test
C01	Apple Trees	129	3905	Healthy Grass	198	1053	Trees	150	23096
C02	Buildings	125	778	Stressed Grass	190	1064	Mostly Grass	150	4120
C03	Ground	105	374	Synthetic Grass	192	505	Mixed Ground Surface	150	6732
C04	Woods	154	8969	Trees	188	1056	Dirt and Sand	150	1676
C05	Vineyard	184	10317	Soil	186	1056	Road	150	6537
C06	Roads	122	3052	Water	182	143	Water	150	316
C07				Residential	196	1072	Shadow of Buildings	150	2083
C08				Commercial	191	1053	Buildings	150	6090
C09				Road	193	1059	Sidewalk	150	1235
C10				Highway	191	1036	Yellow Curb	150	33
C11				Railway	181	1054	Cloth Panels	150	119
C12				Parking Lot 1	192	1041			
C13				Parking Lot 2	184	285			
C14				Tennis Court	181	247			
C15				Running Track	187	473			
	Total	819	29395	Total	2832	12197	Total	1650	52037

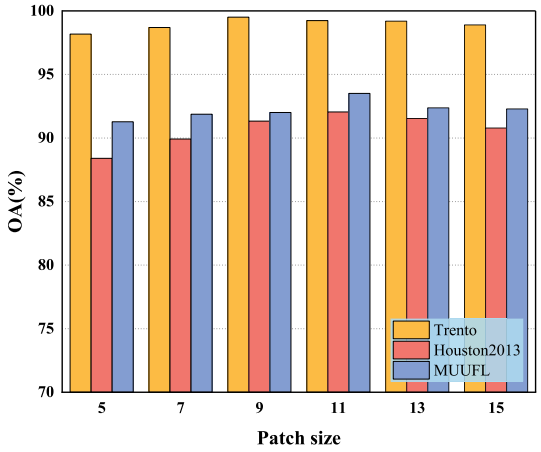


Fig. 4. Impact of patch size on classification performance.

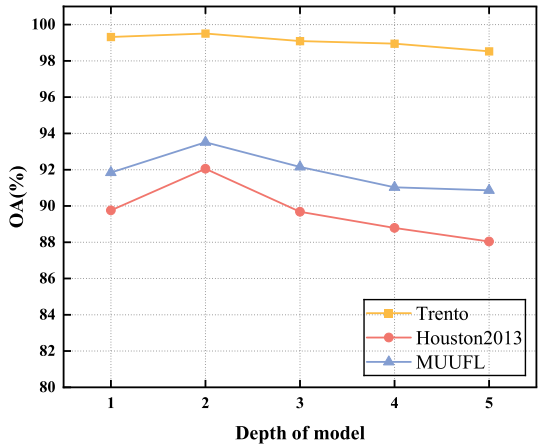


Fig. 5. Impact of depth of model on classification performance.

datasets. The spatial size is 166 × 600 pixels with a spatial resolution of 1 m. The HSI data have a wavelength range of 0.42–0.99 μm and contain 63 spectral bands. In addition, LiDAR data were obtained using the 3100EA sensor. The size of the dataset is 166 × 600 pixels, with a spatial resolution of 1 m. The dataset contains 30 214 marker samples for six land cover types.

2) Houston2013: This dataset consists of scenes collected by the Compact Airborne Spectral Imager (CASI) on the University of Houston campus and neighboring urban areas. It consists of HSI and LiDAR-based digital surface model (DSM) data with a spatial size of 345 × 1905 pixels and a spatial resolution of 2.5 m. The HSI data contain 144 spectral bands with a wavelength range of 0.38–1.35 μm. In addition, the Houston dataset contains a total of 15 029 labeled samples representing 15 different types of land cover classes.

3) MUUFL: The dataset is from the University of Southern Mississippi's Gulfport Campus in Long Beach, Mississippi, USA, and consists of both the HSI and LiDAR datasets. The HSI data has a spatial size of 325 × 220 pixels with 64 spectral channels. Notably, the LiDAR data were captured by the ALTM sensor using a 1064-nm wavelength laser. The dataset contains 53 687 marker samples for 11 land cover types.

Table I lists the names of the land cover classes, the number of training samples, and the number of test samples for the three datasets.

B. Experimental Setup

All the experiments were implemented on the PyTorch platform using an Intel i5 12600KF CPU and an RTX4060Ti GPU. To optimize the network, the Adam optimizer was chosen as the initial optimizer.

To evaluate the classification performance of the proposed framework and other existing models, four widely used quantitative analysis metrics were computed, namely, overall accuracy (OA), average accuracy (AA), Kappa coefficient (κ), and accuracy per class. For each metric, higher values indicate higher classification accuracy. Experiments were conducted on training and test sample sets that were spectrally and spatially disjoint so that there was no interaction between the respective samples.

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TABLE II
CLASSIFICATION PERFORMANCE OF THE TRENTO DATASET OBTAINED BY DIFFERENT METHODS (BEST RESULTS GIVEN IN BOLD)

Class NO.	FusAtNet	S2ENet	MFT	ExViT	M2FNet	GAMF	HLMamba	CHNet	CMFNet
C1	**100.00**	99.64	95.70	99.08	99.00	98.23	96.80	99.15	99.59
C2	**99.96**	98.63	92.98	98.88	99.39	96.26	99.68	98.16	98.13
C3	75.13	81.82	94.92	83.69	91.44	96.52	64.17	97.32	**99.73**
C4	98.93	99.93	99.68	99.87	99.58	99.91	**100.00**	**100.00**	**100.00**
C5	98.86	**100.00**	99.98	99.97	99.87	99.99	99.00	99.54	99.96
C6	89.35	93.05	83.19	93.64	90.82	94.36	89.02	92.50	**97.71**
OA(%)	97.85	98.85	96.85	98.85	98.57	98.75	97.60	98.74	**99.51**
AA(%)	93.71	95.51	94.41	95.86	96.68	97.55	91.45	97.78	**99.17**
$\kappa \times 100$	97.12	98.46	95.78	98.46	98.09	98.33	96.78	98.31	**99.35**

TABLE III
CLASSIFICATION PERFORMANCE OF THE HOUSTON2013 DATASET OBTAINED BY DIFFERENT METHODS (BEST RESULTS GIVEN IN BOLD)

Class NO.	FusAtNet	S2ENet	MFT	ExViT	M2FNet	GAMF	HLMamba	CHNet	CMFNet
C1	79.39	82.15	**90.88**	82.53	83.29	80.53	83.10	83.10	83.10
C2	84.21	**85.15**	84.12	**85.15**	84.40	84.96	83.83	83.18	85.14
C3	89.70	95.84	**99.41**	99.21	97.03	97.23	91.29	98.22	98.02
C4	91.00	92.99	**97.82**	93.94	94.41	91.48	95.93	91.86	92.80
C5	99.34	**100.00**	**100.00**	99.34	99.34	**100.00**	**100.00**	99.91	**100.00**
C6	95.10	**100.00**	**100.00**	**100.00**	95.80	95.80	97.20	**100.00**	**100.00**
C7	91.79	84.05	88.81	89.46	80.69	87.41	85.82	89.18	**94.22**
C8	75.88	**95.25**	77.68	92.12	79.77	74.36	94.11	86.23	91.64
C9	84.80	88.39	78.38	85.84	84.80	80.26	**93.77**	92.92	86.59
C10	61.68	63.61	57.92	66.80	68.24	65.44	67.28	**90.44**	83.49
C11	75.81	93.74	81.02	92.79	96.49	79.22	87.19	92.13	**99.05**
C12	92.89	93.85	89.72	95.39	86.94	89.24	94.72	**96.54**	95.29
C13	72.98	91.93	79.65	86.67	90.18	84.21	**97.72**	84.21	90.80
C14	**100.00**	**100.00**	**100.00**	98.79	99.60	**100.00**	97.57	**100.00**	**100.00**
C15	71.46	96.62	**100.00**	**100.00**	99.58	96.83	90.27	97.25	**100.00**
OA(%)	83.71	89.09	86.25	89.57	87.35	84.93	89.04	91.27	**92.05**
AA(%)	84.40	90.90	88.35	91.20	89.37	87.13	89.99	92.34	**93.35**
$\kappa \times 100$	82.38	88.17	85.12	88.71	86.31	83.76	88.13	90.52	**91.38**

C. Parameter Analysis

In this section, we investigate the impact of two key hyperparameters—patch size (P) and depth of model (N)—on the classification performance of the proposed CMFNet framework.

1) Patch Size: In multisource remote sensing image classification tasks, the selection of patch size plays a pivotal role in network performance by determining the spatial context considered during classification. While larger patches can capture richer contextual information, they simultaneously introduce more noisy pixels. Conversely, smaller patches focus on local details but may lose important spatial correlations. To quantitatively analyze this phenomenon, we conducted comparative experiments on three benchmark datasets with fixed hyperparameters, evaluating candidate patch sizes from the set {5, 7, 9, 11, 13, 15}. As shown in Fig. 4, different patch sizes exhibited varying impacts on OA, with optimal performance achieved at nine for Trento dataset, 11 for both the Houston2013 and MUUFL datasets. This finding demonstrates that medium-sized patches strike an optimal balance between contextual information capture and noise suppression.

2) Depth of Model: We investigate the impact of model depth (ranging from one to five layers) on classification performance using the combined GLFE and CMF layers as fundamental structural units. The experimental results demonstrate that all the three datasets achieve optimal OA at a depth of N = 2, as illustrated in Fig. 5. Further increasing the depth of model leads to degraded classification performance, primarily attributable to overfitting caused by excessive model complexity. These findings conclusively indicate that main-

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TABLE IV
CLASSIFICATION PERFORMANCE OF THE MUUFL DATASET OBTAINED BY DIFFERENT METHODS (BEST RESULTS GIVEN IN BOLD)

Class NO.	FusAtNet	S2ENet	MFT	ExViT	M2FNet	GAMF	HLMamba	CHNet	CMFNet
C1	92.83	92.86	89.38	88.98	91.41	93.50	94.31	90.47	**95.51**
C2	77.77	81.80	81.58	71.55	**87.55**	77.04	83.57	87.06	83.71
C3	84.92	90.06	72.37	79.84	79.77	88.13	79.15	85.71	**91.98**
C4	94.99	**97.79**	94.75	85.44	89.56	96.72	91.27	97.08	94.93
C5	88.27	84.08	83.80	87.32	86.86	89.17	82.87	91.40	**92.39**
C6	99.37	**100.00**	99.37	**100.00**	**100.00**	99.68	99.71	99.05	**100.00**
C7	95.82	97.17	92.17	86.65	94.00	96.35	**97.30**	93.42	94.34
C8	92.22	90.22	95.16	94.74	94.35	**96.37**	92.08	95.27	94.65
C9	84.94	74.82	74.57	78.62	75.87	87.94	**90.83**	83.56	90.69
C10	87.88	**100.00**	87.88	90.91	93.94	96.97	96.83	96.97	**100.00**
C11	98.32	99.16	98.32	**100.00**	94.96	98.32	99.33	99.16	99.16
OA(%)	90.02	90.65	86.55	86.53	89.11	91.43	89.79	90.50	**93.51**
AA(%)	90.66	91.63	88.12	87.65	89.84	92.74	91.57	92.65	**94.30**
$\kappa \times 100$	86.88	87.70	82.49	82.43	85.73	88.71	86.60	87.56	**91.37**

taining an appropriate model depth is crucial for preserving optimal classification performance.

D. Comparative Experiments

To demonstrate the superior performance of the proposed method, this section presents a comprehensive comparative analysis between CMFNet and state-of-the-art approaches—including FusAtNet [53], S2ENet [54], MFT [55], ExViT [56], M2FNet [57], GAMF [58], HLMamba [50], and CHNet [59] from both visualization and quantitative perspectives. For fair comparison, all the competing methods adopt their default parameter settings as reported in their original publications.

1) Quantitative Results and Analysis: Tables II–IV present the comparative classification performance of different methods across three datasets, including evaluation metrics such as OA, AA, κ, and per-class accuracy. The experimental results demonstrate that the proposed CMFNet method significantly outperforms all the existing comparative approaches across every evaluation metric, with the best-performing results highlighted in bold.

Table II presents the classification results on the Trento dataset with CMFNet achieving superior performance as evidenced by its outstanding OA of 99.51%, significantly outperforming all the comparative methods. The network demonstrates particularly strong classification capability for challenging terrain categories, including C3 (Ground) at 99.73% and C6 (Roads) at 97.71%, confirming its effectiveness in handling complex features through cross-modal fusion. In contrast, MFT and HLMamba show relatively weaker performance across most categories, especially for C3 (Ground) and C6 (Roads) classification, which may be attributed to their less effective global feature extraction. While ExViT and S2ENet achieve comparable accuracy in specific categories such as C1 (Apple Trees) and C4 (Woods), their overall generalization capability remains inferior to CMFNet's consistently superior performance across all the metrics.

Table III presents the quantitative results on the Houston2013 dataset, where CMFNet demonstrates superior performance, achieving OA of 92.05%, AA of 93.35%, and κ of 91.38%, owing to its innovative CMF architecture. The model effectively captures both spectral and geometric features, with notable accuracies of 100% for class C5 (Soil) and C6 (Water). Its progressive fusion mechanism adeptly addresses challenging cases, as reflected by the 99.05% accuracy achieved on C11 (Railway). Furthermore, compared with conventional approaches, the GLFE module within CMFNet delivers enhanced performance in complex urban environments, attaining perfect classification (100%) for classes C14 (Tennis Court) and C15 (Running Track). These results collectively validate the effectiveness of CMFNet in joint multimodal feature representation and in significantly improving classification accuracy.

Table IV presents the classification performance of various methods on the MUUFL dataset. The outstanding results of CMFNet (OA 93.16%, AA 93.95%, κ 90.93%) fully validate the effectiveness of its CMF architecture. For categories requiring combined geometric and spectral features like C3 (Mixed Ground Surface) and C5 (Road), CMFNet achieves significantly higher accuracy (91.98% and 92.39%, respectively) than other methods, benefiting from the GLFE module's synergistic modeling of local details and global context. While performance in some categories is comparable to other top methods, CMFNet's overall superiority—particularly its 90.69% accuracy for C9 (Sidewalk) and perfect 100% for C10 (Yellow Curb)—demonstrates the robustness of its hierarchical feature fusion strategy.

2) Visual Evaluation and Analysis: Figs. 6–8 present comparative visualization results of different methods. Qualitative evaluation demonstrates that the proposed CMFNet generates the most accurate classification maps with minimal noise

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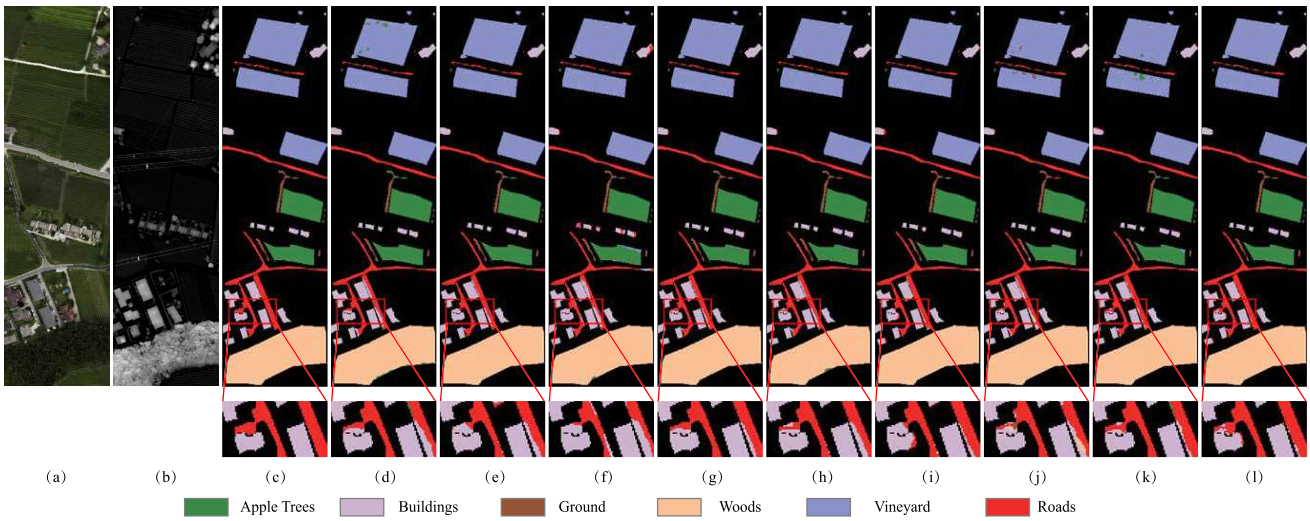


Fig. 6. Classification maps of the Trento dataset obtained by different methods. (a) Pseudocolor map of HSI. (b) LiDAR image. (c) Ground truth. (d) FusAtNet. (e) S2ENet. (f) MFT. (g) ExViT. (h) M2FNet. (i) GAMF. (j) HLMamba. (k) CHNet. (l) CMFNet.

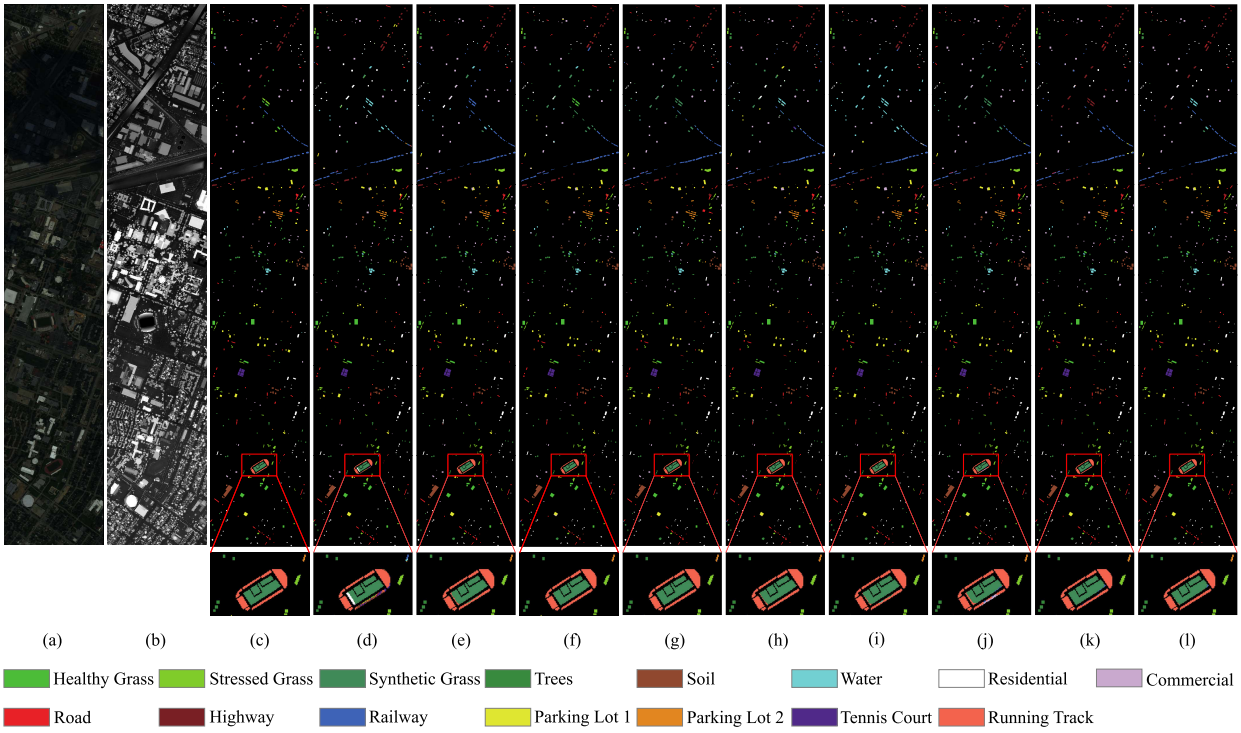


Fig. 7. Classification maps of the Houston2013 dataset obtained by different methods. (a) Pseudocolor map of HSI. (b) LiDAR image. (c) Ground truth. (d) FusAtNet. (e) S2ENet. (f) MFT. (g) ExViT. (h) M2FNet. (i) GAMF. (j) HLMamba. (k) CHNet. (l) CMFNet.

interference, visually validating the quantitative conclusions. The visual comparisons clearly reveal performance differences in classification accuracy and noise suppression across methods. Notably, CMFNet exhibits superior performance in both boundary preservation and homogeneous region consistency.

The visual analysis reveals that the classification maps generated by CMFNet exhibit remarkable boundary sharpness and high purity within homogeneous regions, a direct result of its innovative progressive dual-branch fusion architecture. Specifically, the method demonstrates outstanding performance across three benchmark datasets. On the Trento dataset, CMFNet leverages multiscale feature embedding to simultaneously preserve fine spatial structures and high-level semantic features, enabling precise recognition of geometric details such as roads and buildings. In the Houston2013 dataset, the dynamic interaction mechanism implemented by the CMF module enhances feature complementarity, particularly in complex scenes such as heterogeneous land covers. For the MUUFL dataset, the GLFE module effectively integrates global context and local features, significantly improving the accuracy of identifying small-scale targets such as yellow curbs, while also suppressing salt-and-pepper noise in the classification maps. The visual results, supported by quantitative metrics, not only illustrate CMFNet's superior capabilities

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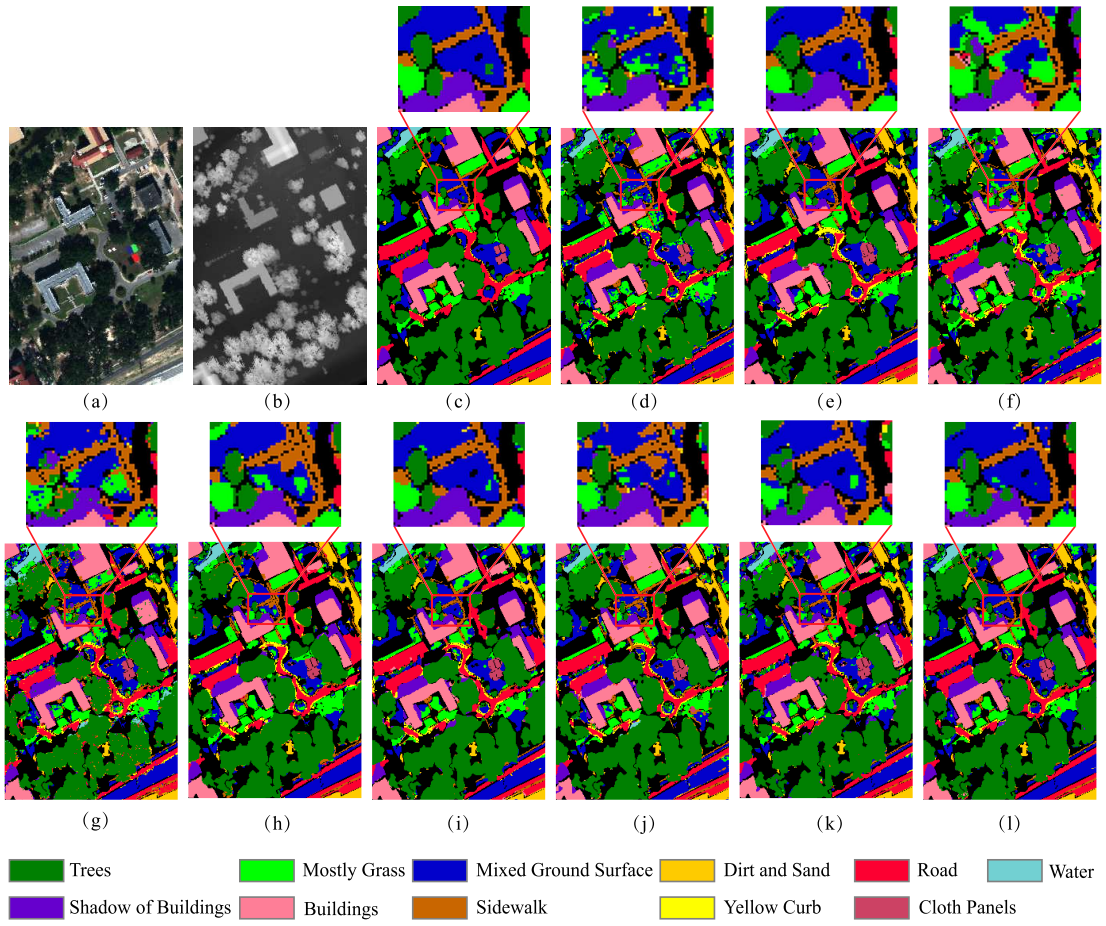


Fig. 8. Classification maps of the MUUFL dataset obtained by different methods. (a) Pseudocolor map of HSI. (b) LiDAR image. (c) Ground truth. (d) FusAtNet. (e) S2ENet. (f) MFT. (g) ExViT. (h) M2FNet. (i) GAMF. (j) HLMamba. (k) CHNet. (l) CMFNet.

TABLE V
MODEL COMPLEXITY COMPARISON ACROSS DIFFERENT METHODS AND DATASETS

Dataset	Complexity	FusAtNet	S2ENet	MFT	ExViT	M2FNet	HLMamba	GAMF	CHNet	CMFNet
Trento	Parameters	36.24M	177.26K	263.53K	226.41K	189.35K	198.05K	3.99M	20.77M	422.53K
	FLOPs	3.38G	21.48M	12.71M	32.58M	76.06M	57.53M	1.31G	160.83M	36.35M
	Inference time	16.89s	1.67s	1.97s	5.36s	182.49s	5.56s	22.32s	10.70s	5.92s
Houston2013	Parameters	36.90M	270.87K	313.07K	229.01K	255.35K	221.37K	7.31M	20.82M	484.75K
	FLOPs	3.46G	32.77M	25.30M	32.82M	90.64M	78.43M	3.09G	160.88M	65.52M
	Inference time	7.47s	0.78s	1.18s	2.56s	77.15s	2.92s	17.06s	4.51s	3.29s
MUUFL	Parameters	36.25M	178.48K	249.39K	226.46K	127.09K	198.65K	3.21M	20.79M	356.81K
	FLOPs	3.39G	21.62M	11.04M	32.58M	68.57M	58.07M	1.23G	160.86M	46.31M
	Inference time	30.56s	3.02s	3.92s	9.85s	373.04s	9.28s	40.33s	31.08s	9.93s

in boundary preservation, small-object recognition, and scene adaptability but also visually validate the effectiveness of the CMF mechanism in enhancing classification accuracy for multimodal remote sensing data.

3) Computational Complexity Analysis: As shown in Table V, model complexity plays a crucial role in evaluating the computational efficiency and resource utilization of deep learning methods. CMFNet achieves an optimal balance among parameter size, computational cost, and inference speed. Its parameter count ranges from 356.81K to 484.75K, which is slightly higher than that of the MFT model (249.39K–313.07K), yet far smaller than performance-comparable high-complexity models such as GAMF, with 3.21M–7.31M parameters, and FusAtNet, with 36.24M–36.90M parameters—amounting to only one-tenth or even one-hundredth of their parameter size while delivering comparable or superior performance. In terms of computational complexity, CMFNet's FLOPs range from 36.35M to 65.52M, representing a reduction of several orders of magnitude compared with FusAtNet (3.38G–3.46G) and GAMF (1.23G–3.09G), while still achieving about 10%–20% savings over medium-complexity models such as HLMamba (57.53M–78.43M FLOPs). Regarding inference time, CMFNet also demonstrates excellent performance, with runtime ranging

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TABLE VI
PERFORMANCE COMPARISON USING DIFFERENT DATA MODALITIES (BEST RESULTS GIVEN IN BOLD)

Case	Trento			Houston2013			MUUFL		
	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
Only LiDAR	93.19	84.98	90.22	63.21	61.52	60.32	80.41	79.18	74.86
Only HSI	97.68	95.33	96.70	87.90	89.88	36.91	90.45	90.65	87.11
HSI+LiDAR	**99.51**	**99.17**	**99.35**	**92.05**	**93.35**	**91.38**	**93.51**	**94.30**	**91.37**

TABLE VII
ABLATION STUDY OF CMFNET COMPONENTS ON THREE DATASETS (BEST RESULTS GIVEN IN BOLD)

Case	MFEncoder		GLFE			CMF	Trento			Houston2013			MUUFL		
	3Dconv	Multi2D	Global	Local	AFF		OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
1	✓	Conv2D	×	×	×	×	97.52	94.20	96.68	86.44	88.76	85.29	85.19	85.61	80.74
2	✓	✓	×	×	×	×	98.39	94.05	97.83	87.58	89.74	86.51	87.81	88.10	84.03
3	✓	✓	✓	×	×	×	98.68	95.80	98.24	88.40	89.96	87.46	91.05	92.56	87.96
4	✓	✓	×	✓	×	×	98.74	96.13	98.78	88.76	90.42	87.84	91.25	92.86	88.02
5	✓	✓	✓	✓	×	×	99.02	98.21	98.05	90.62	91.58	89.74	92.05	93.01	89.56
6	✓	✓	✓	✓	✓	×	99.27	98.36	99.02	90.65	91.79	89.87	92.44	93.23	90.02
7	✓	✓	✓	✓	✓	✓	**99.51**	**99.17**	**99.35**	**92.05**	**93.35**	**91.38**	**93.51**	**94.30**	**91.37**

from 3.29 to 9.93 s across the three datasets—consistently outperforming GAMF (17.06–40.33 s) and FusAtNet (7.47–30.56 s)—and requiring only 9.93 s on the MUUFL dataset, far less than M2FNet's 373.04 s and even lower than HLMamba's 9.28 s. On the Trento dataset, it achieves 5.92 s inference, roughly one-quarter of GAMF and one-third of FusAtNet. These results show that CMFNet strategically accepts a moderate increase in parameter size to gain substantial advantages in computational efficiency and inference speed across all three datasets, while maintaining optimal or near-optimal classification accuracy, underscoring its strong practical value in resource-constrained remote sensing applications that demand high precision.

E. Ablation Analysis

To ensure fairness and reproducibility, all the ablation experiments were conducted under consistent training settings. Specifically, the number of training epochs was set to 200, the initial learning rate was 1e-4, and the batch size was 64. Other training parameters remained the same as those used in the main model.

1) Ablation Analysis of Different Modal Data Inputs: The ablation analysis of CMFNet's multimodal configurations clearly demonstrates the distinct contribution patterns of different data modalities to classification performance. Table VI demonstrates the classification performance of different modality inputs across three datasets. The experimental results reveal that when using LiDAR data alone, the model achieves OAs of 93.19%, 63.21%, and 80.41% on the Trento, Houston2013, and MUUFL datasets, respectively. Switching to HSI-only input elevates these metrics to 97.68%, 87.90%, and 90.45%, confirming HSI data's superior discriminative power across all the test scenarios. The complete CMFNet framework, leveraging fused HSI and LiDAR data, achieves optimal performance with overall accuracies of 99.51%, 92.05%, and 93.51%, accompanied by enhanced κ of 99.35%, 91.38%, and 91.37% across the three benchmark datasets. These results provide definitive validation of the CMF's effectiveness in excavating and synthesizing complementary information: the precise 3-D geometric features from LiDAR synergistically combine with HSI's rich spectral signatures to substantially enhance classification capability in complex scenarios. The ablation outcomes offer compelling evidence for the value of our proposed multimodal fusion strategy, demonstrating that systematic cross-modal feature interaction can successfully transcend the performance limitations inherent in single-modality approaches.

2) Ablation Analysis of Different Components: The ablation study on CMFNet's architectural components systematically evaluates each module's contribution to model performance. The experimental results in Table VII show consistent performance improvements across all the three datasets with progressive module integration. The baseline architecture (Case1) using conventional 3-D convolution and standard 2-D convolution demonstrates competent performance (97.52% OA on Trento). The incorporation of multiscale 2-D convolution (Case2) produces substantial performance gains, most notably increasing MUUFL's OA from 85.19% to 87.81%, conclusively demonstrating the critical role of multiscale feature extraction. The GLFE module (Case3) produces a qualitative leap in performance, with Trento's OA jumping from 98.39% to 98.68%, confirming the global context's critical role in complex scene understanding. Moreover, the local feature extraction module (Case4) demonstrated robust capability in capturing fine-grained details, achieving an OA of 98.74% on the Trento dataset. The incorporation of the AFF (Case6) module significantly enhances model performance through the effective integration of global and local features. On the Trento dataset, the AFF module yields notable improvements of 0.25% in OA, 0.15% in AA, and 0.97% in κ compared with the baseline without AFF (Case5). The most significant performance enhancement is achieved through the CMF module (Case7), which establishes state-of-the-art results across all the three benchmark datasets: achieving 99.51% OA on Trento, 92.05% on Houston2013, and 93.51% on MUUFL.

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This graduated performance enhancement clearly illustrates each module's design value while verifying the technical rationale behind CMFNet's MFEncoder, GLFE, and CMF architecture.

IV. Conclusion

This article proposes CMFNet for hyperspectral and LiDAR data classification, featuring an innovative dual-branch architecture that effectively extracts and fuses multimodal features. The proposed framework integrates three key components: 1) a GLFE module that combines CNN and Mamba networks to achieve adaptive fusion of global and local representations; 2) a novel CMF module that extends the Mamba architecture to enhance dual-modal feature interaction; and 3) a hierarchical multilayer structure designed to fully exploit the complementary information between HSI and LiDAR data. Extensive experiments conducted on three benchmark remote sensing datasets demonstrate that CMFNet significantly outperforms state-of-the-art methods in terms of both classification accuracy and computational efficiency. Ablation studies further validate the effectiveness of each component in the proposed framework.

Despite these promising results, some limitations remain. The current model has not yet been extensively validated in large-scale or more complex real-world scenarios, where multimodal data may vary in quality, resolution, or alignment. Moreover, the reliance on fully supervised learning requires substantial labeled data, which can be costly and labor-intensive to obtain. In future work, we plan to enhance the adaptability of CMFNet in more diverse and challenging environments, develop more efficient and lightweight cross-modal fusion mechanisms, and incorporate self-supervised or weakly supervised learning strategies to alleviate dependence on manual annotations. These efforts aim to further improve the accuracy, scalability, and practicality of multimodal remote sensing data analysis.

References

[1] Y. Sun, Z. Fu, C. Sun, Y. Hu, and S. Zhang, "Deep multimodal fusion network for semantic segmentation using remote sensing image and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404418.

[2] H. Yu et al., "Global spatial and local spectral similarity-based manifold learning group sparse representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3043–3056, May 2020.

[3] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[4] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.

[5] X. Wu, D. Hong, and J. Chanussot, "Convolutional neural networks for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517010.

[6] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911.

[7] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

[8] C. Weber et al., "Hyperspectral imagery for environmental urban planning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 1628–1631.

[9] P. Wang, Z. He, B. Huang, M. Dalla Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 5, pp. 9268–9282, May 2025.

[10] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[11] P. Wang et al., "Low-rank tensor completion pansharpening based on haze correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5405720.

[12] L. Karthikeyan, I. Chawla, and A. K. Mishra, "A review of remote sensing applications in agriculture for food security: Crop growth and yield, irrigation, and crop losses," *J. Hydrol.*, vol. 586, Jul. 2020, Art. no. 124905.

[13] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[14] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[15] Z. Wang and M. Menenti, "Challenges and opportunities in LiDAR remote sensing," *Frontiers Remote Sens.*, vol. 2, Mar. 2021, Art. no. 641723.

[16] A. Kuras, M. Brell, J. Rizzi, and I. Burud, "Hyperspectral and LiDAR data applied to the urban land cover machine learning and neural-network-based classification: A review," *Remote Sens.*, vol. 13, no. 17, p. 3393, Aug. 2021.

[17] S. Jia, X. Zhou, S. Jiang, and R. He, "Collaborative contrastive learning for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507714.

[18] J. Lin, F. Gao, X. Shi, J. Dong, and Q. Du, "SS-MAE: Spatial–spectral masked autoencoder for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531614.

[19] S. Xia, X. Zhang, H. Meng, and L. Jiao, "Ternary modality contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522017.

[20] C. Shi, T. Wang, and L. Wang, "Branch feature fusion convolution network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5194–5210, 2020.

[21] W. Han, W. Miao, J. Geng, and W. Jiang, "CMSE: Cross-modal semantic enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509814.

[22] S. K. Roy, A. Deria, D. Hong, M. Ahmad, A. Plaza, and J. Chanussot, "Hyperspectral and LiDAR data classification using joint CNNs and morphological feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530416.

[23] M. Zhang, W. Li, Q. Du, L. Gao, and B. Zhang, "Feature extraction for classification of hyperspectral and LiDAR data using patch-to-patch CNN," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 100–111, Jan. 2020.

[24] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[25] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[26] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[27] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[28] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[29] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[30] F. Zhu, C. Shi, K. Shi, and L. Wang, "Joint classification of hyperspectral and LiDAR data using hierarchical multimodal feature aggregation-based multihead axial attention transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503817.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:56:04 UTC from IEEE Xplore. Restrictions apply.

[31] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021.
[32] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art. no. 5518615.
[33] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, Jan. 2022, Art. no. 5522214.
[34] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615.
[35] J. X. Yang, J. Zhou, J. Wang, H. Tian, and A. W.-C. Liew, "LiDAR-guided cross-attention fusion for hyperspectral band selection and image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515815.
[36] T. Song, Z. Zeng, C. Gao, H. Chen, and J. Li, "Joint classification of hyperspectral and LiDAR data using height information guided hierarchical fusion-and-separation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505315.
[37] H. Chang, H. Bi, F. Li, C. Xu, J. Chanussot, and D. Hong, "Deep symmetric fusion transformer for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644115.
[38] Z. Li, K. Zheng, L. Gao, N. Zi, and C. Li, "Feature reconstruction guided fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4408914.
[39] F. Gao et al., "Prototype-based information compensation network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513615.
[40] X. Wang, L. Song, Y. Feng, and J. Zhu, "S^3F^2Net: Spatial–spectral-structural feature fusion network for hyperspectral image and LiDAR data classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4801–4815, May 2025.
[41] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5470–5483, 2024.
[42] X. Wei, W. Yu, H. Huang, L. Sun, C. Zhao, and G. Shen, "MAMInet II: Illumination-insensitive modalitywise assimilation guided multistage interaction network for hyperspectral and LiDAR joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539513.
[43] W. Yu, L. Gao, H. Huang, Y. Shen, and G. Shen, "HI^2D^2FNet: Hyperspectral intrinsic image decomposition guided data fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521715.
[44] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[45] Y. Liu et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.
[46] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216.
[47] Z. Pan, C. Li, A. Plaza, J. Chanussot, and D. Hong, "Hyperspectral image classification with Mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602814.
[48] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.
[49] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-Mamba: Interval group spatial–spectral Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817.
[50] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915.
[51] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multisource remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504116.
[52] M. Cao et al., "M^3amba: CLIP-driven mamba model for multi-modal remote sensing classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 8, pp. 7605–7617, Aug. 2025.
[53] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 92–93.
[54] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[55] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.
[56] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[57] L. Sun, X. Wang, Y. Zheng, Z. Wu, and L. Fu, "Multiscale 3-D–2-D mixed CNN and lightweight attention-free transformer for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2100116.
[58] J. Cai et al., "A novel graph-attention based multimodal fusion network for joint classification of hyperspectral image and LiDAR data," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123587.
[59] K. Ni, Y. Xie, G. Zhao, Z. Zheng, P. Wang, and T. Lu, "Coarse-to-fine high-order network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5509716.



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